# Yusef Syed

**Security Engineer, AI agents & evals | 14 merged PRs in 6 repos, incl. Microsoft, W&B, Apache Arrow**

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Toronto, ON | U.S. & Canadian citizen, no sponsorship | **Available full-time from January 2027,** SF or NYC

## Education

**University of Toronto** **Expected May 2030**

*B.Sc., Mathematical & Physical Sciences (first year); intended Math and CS double major*

## Open-Source Contributions

**Six repos:** PyRIT, Agent Lightning, W&B, Inspect Scout, Arrow, Kornia | Full list **Sep – Oct 2026**

- Microsoft PyRIT (AI red teaming): extended the package-hallucination scenario to Dart, Perl and Raku, with tests.
- Microsoft Agent Lightning (4 merged PRs): reliability fixes, e.g. case-insensitive auth-header replacement.
- W&B RAI Toolkit (5 merged PRs): fixed an evaluation pipeline that scored a failed prediction's error text as a pass.
- Inspect Scout (2 merged PRs): per-item model-usage reporting for custom loaders; a scanner-loading fix.
- Apache Arrow: extended four PyArrow slice APIs to accept Arrow integer scalars, with regression tests.

## Experience

**Scopehaven | github.com/YusefSyed/scopehaven-oss** **Oct 2026 – Present**

*Founder and software engineer, solo startup; security testing for AI agents*

- Built **Scopehaven**, a Node.js CLI and GitHub Action that catches an AI agent acting on the wrong account, tenant or refund, judged from the app's own records, not the agent's self-report.
- Released the **Apache-2.0 open edition** (v0.7.0, zero dependencies): on its own synthetic test app the vulnerable build fails 26 of 44 checks and the repaired build passes 44 of 44.
- Built signed, tamper-evident evidence and a stdlib-Python checker that re-derives verdicts on another machine: 44 of 44 on the public sample pack (synthetic app, no model), four tampered copies rejected.
- Filed 15 private security advisories and 2 HackerOne reports (authorization, credential and data-exposure flaws) with 10 open-source projects, each reproduced 3 of 3 by an automated harness; all awaiting triage.

## Preprints

**ToolEffectBench: Scoring Prompt Injection on Tool-Using Agents from Persisted State** **2026**

*Zenodo, doi:10.5281/zenodo.23966727 | arXiv: in review | code & logs*

- Built **ToolEffectBench**, an Inspect AI benchmark that scores prompt injection by what the agent changed in a network-less Docker sandbox, not by its transcript (17 models, 5,628 scored runs).
- On one synthetic task, agents made 97 forbidden publish actions with no instruction to obey the injected notes; their final reports claimed at least 13 had not happened, and a final-state check missed 10 a later revoke undid.

**Checks for Checkers: Sealed Paired Comparisons of AI Agent Verification Tools** **2026**

*Zenodo, doi:10.5281/zenodo.23966747 | arXiv: in review*

- Blind, timestamped trial on 96 planted-defect and 96 clean cases: my own tool caught 83, an open-source eval framework 88, neither with a false alarm; not a significant gap ($p = 0.30$); published as a null result.

## Projects

**effect-tracer** | *Rust, Linux ptrace/seccomp, Ed25519* | github.com/YusefSyed/effect-tracer **2026**

- Built a tracer that signs a process tree's file writes, connections, DNS lookups and execs into a hash chain; in seccomp mode: 19 of 20 seeded bypasses caught, 1.1–1.9x overhead on four workloads, 0 undeclared events in 30 clean runs.

## Technical Skills

**Languages:** Python, TypeScript, JavaScript, SQL, Rust (effect-tracer)
**Security & evals:** AI security, AI red teaming, prompt-injection evals, LLM agents, application security (AppSec), access-control testing (IDOR, row-level security), vulnerability research, responsible disclosure
**Tools:** Inspect AI, PyRIT, Linux (ptrace, seccomp), Node.js, MCP, PostgreSQL, GitHub Actions (CI/CD)

## Awards & Activities

**Providence** (Swift, team of 4): **1st Place, Aramco Americas Best Beginner Hack,** Hack the North 2026
**USA Powerlifting (60 kg, age 16–17, 2025):** U.S. #20 | **Feeding the Valley,** VP: $20,000 raised for food banks